



MERIDIAN JUNIOR COLLEGE
2009 JC2 PRELIMINARY EXAMINATION

MATHEMATICS
Higher 2
Paper 1

9740/01

Tuesday

15 September 2009
3 hours

Additional materials: Writing paper
List of Formulae (MF 15)

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READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

- 1 Express $\frac{2}{r^2 + 6r + 8}$ in partial fractions. [2]

Find

$$\sum_{r=1}^n \frac{2}{r^2 + 6r + 8},$$

giving your answer in the form $k - f(n)$, where k is a constant. [2]

Find also $\sum_{r=1}^n \frac{r^2 + 6r + 88}{r^2 + 6r + 8}$. [2]

- 2 It is given that $x_1, x_2, x_3, \dots$ is a sequence of positive numbers, where $x_1 = 3$ and $x_{n+1} = x_n^2 - 3x_n + 4$, for $n \in \mathbb{N}^+$.

Given also that $S_n = \sum_{i=1}^n \frac{1}{x_i - 1}$, prove by induction, for all $n \in \mathbb{N}^+$,

$$S_n = 1 - \frac{1}{x_{n+1} - 2}. \quad [6]$$

- 3 Using the substitution $x = 2e^t$, show that

$$\int \frac{\ln x - \ln 2}{x\sqrt{\ln x - \ln 2 - 2}} dx = \int \left[\sqrt{t - 2} + \frac{2}{\sqrt{t - 2}} \right] dt.$$

Hence evaluate $\int_{2e^2}^{2e^4} \frac{\ln x - \ln 2}{x\sqrt{\ln x - \ln 2 - 2}} dx$, giving your answer in the form $\frac{a\sqrt{2}}{b}$

where $a, b \in \mathbb{N}$. [7]

- 4** ToTo bear is an endangered species on Too Island. The number of births per unit time is proportional to the population of ToTo bears at any time and the number of deaths per unit time is proportional to the square of the population of ToTo bears.

If the population at time t is x , show that $\frac{dx}{dt} = ax - bx^2$, where a and b are positive constants.

It is also known that the population of ToTo bears remain constant when $x = 200$.

Hence, show that $x = \frac{200}{Be^{-kt} + 1}$, where k is a positive constant.

Deduce that there is a limit to the size of the population of ToTo bears and explain the significance of this limit. [7]

- 5** With respect to an origin O , the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are not parallel.

(i) It is given that B lies on the line segment AC , such that $\overrightarrow{BC} = 2\mathbf{b} - k\mathbf{a}$. State the value of the constant k . Hence, find $\overrightarrow{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) $\overrightarrow{ON}$ is the projection vector of $\overrightarrow{OB}$ onto $\overrightarrow{OA}$, where $\overrightarrow{ON} = \frac{3}{4}\mathbf{a}$ and

$\mathbf{a} \cdot \mathbf{b} = 4$. Show that the length of OA is $\frac{4\sqrt{3}}{3}$. [2]

(iii) Given also that $\mathbf{b} = \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix}$, find the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$. [2]

(iv) Given that $OADC$ is a parallelogram and E lies on line segment AC such that $AE : AC = 2 : 3$, find $\overrightarrow{DE}$. [3]

- 6 The complex number z satisfies the relations $|z-1|=1$ and $\arg(z-2)=\frac{7\pi}{8}$.
- (i) Illustrate both of these relations on a single Argand diagram. [2]
 - (ii) Find the exact value of z in the form $a+ib$. [3]
 - (iii) Hence, deduce the exact value of $\tan\frac{3\pi}{8}$. [2]
 - (iv) Find the minimum value of $\arg(z-1-i)$ in exact form, given that $|z-1|\leq 1$ and $\arg(z-2)=\frac{7\pi}{8}$. [2]
- 7 A graphic calculator is **not** to be used in answering this question.
- (a) Express $(1-4i)^2$ in the form $a+bi$. Hence, find the roots of the equation
$$\left(\frac{z}{2}+3\right)^2=15+8i. \quad [4]$$
 - (b) Find the exact roots of the equation $(z-2)^3=8$ in the form $re^{i\theta}$, where $r>0$ is non-trigonometrical.
Hence, find the roots of the equation
$$\left(\frac{1-2z}{z}\right)^3=8. \quad [6]$$
- 8
- (a) Find the series expansion of $(1+e^x)^n$ in ascending powers of x , up to and including the term in x^2 . [4]
 - (b)
 - (i) Expand $y=\frac{x}{3-2x}$ in ascending powers of x , up to and including the term in x^3 . [3]
 - (ii) State the set of values of x for which the expansion is valid. [1]
 - (iii) By considering the coefficient of x^n in this expansion for y , find $\frac{d^n y}{dx^n}$ at $x=0$. [2]

- 9** The equations of three planes Π_1 , Π_2 , and Π_3 are

$$\begin{aligned}x + 3y + az &= 8, \\ 3x + y + bz &= 0 \quad \text{and} \\ x + py + qz &= 11,\end{aligned}$$

respectively, where a , b , p and q are constants.

- (i)** If the point $(\alpha, \beta, 0)$ lies on both Π_1 and Π_2 , find the values of α and β .
[2]

- (ii)** Given that the line l_1 lies on both Π_1 and Π_2 , find a vector equation of l_1 in terms of a and b .
[2]

The line l_2 has equation $\mathbf{r} = \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix}$, where μ is a parameter.

- (iii)** Given that l_1 and l_2 intersect at a point and $a + b = 0$, find the values of a and b .
[4]
- (iv)** If Π_1 , Π_2 , and Π_3 have no point in common, using the values of a and b found in **(iii)**, comment on the value of p and find an equation relating p and q .
[3]

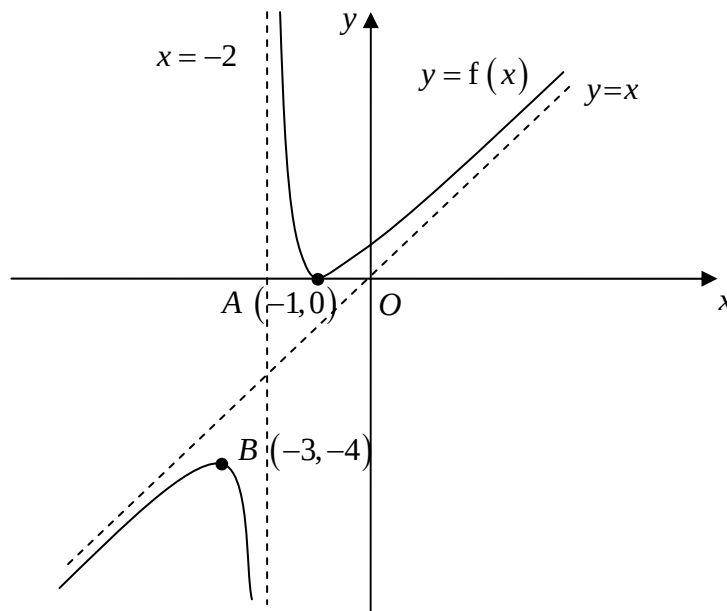
- 10 (a)** A geometric progression and an arithmetic progression has a common first term, a .

The sum of the first three terms of the geometric progression is $\frac{26}{27}$ of the sum to infinity. The second and third terms of the geometric progression are the n th and $(n+1)$ th terms of the arithmetic progression respectively. Find the sum of the first 55 terms of the arithmetic progression in terms of a .
[5]

- (b) Ann is learning to drive. Her first lesson is 30 minutes long. Her second lesson is 35 minutes long. Each subsequent lesson is 5 minutes longer than the lesson before.

- (i) How long would Ann's seventh lesson be? [1]
 (ii) How many hours of lesson would Ann have completed after her seventh lesson? [2]
 (iii) During which lesson would Ann have completed a total of 60 hours of driving lessons? [3]

- 11 (a) The diagram shows the graph of $y = f(x)$. The equations of the asymptotes, also shown in the diagram, are $x = -2$ and $y = x$.



Sketch the graph of $y = f'(x)$, indicating clearly the asymptote(s) and the coordinates of the point corresponding to A and B. [3]

- (b) A curve has parametric equations

$$x = a \cos t, \quad y = a \sin t, \quad \text{for } 0 < t \leq \pi,$$

where a is a positive constant.

The tangent to the curve at the point where $x = \frac{a}{2}$, meets the x - and y - axes at Q and R respectively. The origin is denoted by O .

- (i) Show that the equation of tangent to the curve at $x = \frac{a}{2}$ is

$$3y = \sqrt{3}(2a - x) \quad [2]$$

- (ii) Hence, find the exact area of triangle OQR in terms of a . [3]

- (c) A manufacturer wants to design a lidless container with a square base and a capacity of 1000 cm^3 . Given that the uniform thickness of the container is 1 cm, using differentiation, find the internal length of the tank, x cm, such that the volume of the material used is minimised.

[6]

